

AERIAL ROBOTICS LOCOMOTION

AZMI BIN BASHARUDIN

2211387

Key Design Issues



- Mobility Mechanism: Wheeled (UGVs), legged, tracked, aerial (rotor), underwater (propeller).
- Stability: How well it can handle different terrains or flight conditions.
- Weight Distribution: Affects balance, energy consumption, and payload.
- Actuation Method: Electric, hydraulic, or pneumatic actuators.



Attributes



- Speed and Range: Based on mission requirements and energy capacity.
- Endurance: Time it can function without recharging/refueling.
- Terrain Adaptability: How well it can navigate uneven or unknown terrain.



Basic Theoretical Model

- Kinematics: Motion of parts without considering forces.
- Dynamics: Interaction of forces and motion; critical for control.
- Control Systems: PID controllers, fuzzy logic, neural networks for stability and path tracking.



Payload



Payload capacity depends on size and application:

- Surveillance: Camera systems, GPS, LiDAR.
- Delivery: Small packages.
- Agriculture: Spraying equipment.
- Military: Communication jammers or explosive neutralizers.



Thank You!

